



Figure 1: images/aa841dc166a15009c1f28d283b4f0f0b34a7bf0b3958a2b8e7dec46b4a7b86a3.jpg

CoSToM: Causal-oriented Steering for Intrinsic Theory-of-Mind Alignment in Large Language Models

CoSToM: 基于因果导向的内在心智理论对齐方法在大型语言模型中的应用

Mengfan Li ^{1*}, Xuanhua Shi ^{1†}, Yang Deng ²

李梦凡 ^{1*}, 石萱华 ^{1†}, 杨登 ²

¹ National Engineering Research Center for Big Data Technology and System, Services Computing Technology and System Lab, Cluster and Grid Computing Lab, Huazhong University of Science and Technology

¹ 国家大数据技术与系统工程技术研究中心, 服务计算技术与系统实验室, 集群与网格计算实验室, 华中科技大学

² Singapore Management University

² 新加坡管理大学

{limf, xhshi}@hust.edu.cn, ydeng@smu.edu.sg

{limf, xhshi}@hust.edu.cn, ydeng@smu.edu.sg

Abstract

摘要

Theory of Mind (ToM), the ability to attribute mental states to others, is a hallmark of social intelligence. While large language models (LLMs) demonstrate promising performance on standard ToM benchmarks, we observe that they often fail to generalize to complex task-specific scenarios, relying heavily on prompt scaffolding to mimic reasoning. The critical misalignment between the internal knowledge and external behavior raises a fundamental question: Do LLMs truly possess intrinsic cognition, and can they externalize this internal knowledge into stable, high-quality behaviors? To answer this, we introduce CoSToM ¹ (Causal-oriented Steering for ToM alignment), a framework that transitions from mechanistic interpretation to active intervention. First, we employ causal tracing to map the internal distribution of ToM features, empirically uncovering the internal layers' characteristics in encoding fundamental ToM semantics. Building on this insight, we implement a lightweight alignment framework via targeted activation steering within these ToM-critical layers. Experiments demonstrate that CoSToM significantly enhances human-like social reasoning capabilities and downstream dialogue quality.

心智理论 (Theory of Mind, 简称 ToM) 是指将心理状态归因于他人的能力, 是社会智能的标志。尽管大型语言模型 (LLMs) 在标准的 ToM 基准测试中表现出色, 但我们观察到它们常常难以推广到复杂的任务特定场景, 往往依赖于提示模板来模拟推理。内部知识与外部行为之间的关键不匹配引发了一个根本性问题: LLMs 是否真正具备内在认知能力, 能否将这种内部知识外部化为稳定、高质量的行为? 为了解答

这一问题，我们引入了 CoSToM¹（面向因果的 ToM 对齐引导），一个从机制性解释转向主动干预的框架。首先，我们采用因果追踪来映射 ToM 特征的内部分布，实证地揭示了编码基本 ToM 语义的内部层的特性。基于这一洞察，我们在这些关键的 ToM 层中通过有针对性的激活引导实现了一个轻量级的对齐框架。实验表明，CoSToM 显著提升了类人社会推理能力及下游对话质量。

1 Introduction

1 引言

Theory of Mind (ToM), the inherent ability to attribute mental states such as beliefs, desires, and intentions to others, stands as a hallmark of human social intelligence (Baker et al., 2017; Strachan et al., 2024). It enables individuals to anticipate others' motives, knowledge states, and reactions, and thus forms the cognitive basis of complex social communication, such as persuasion (Wang et al.; Mishra et al., 2022; Tiwari et al., 2022), negotiation (Deng et al., 2023b; Zhan et al., 2024; Kwon et al.,

心智理论 (Theory of Mind, 简称 ToM) 是人类固有的能力，能够将信念、欲望和意图等心理状态归因于他人，是人类社会智能的标志 (Baker 等, 2017; Strachan 等, 2024)。它使个体能够预测他人的动机、知识状态和反应，从而构成了复杂社会交流的认知基础，例如说服 (Wang 等; Mishra 等, 2022; Tiwari 等, 2022)，谈判 (Deng 等, 2023b; Zhan 等, 2024; Kwon 等,

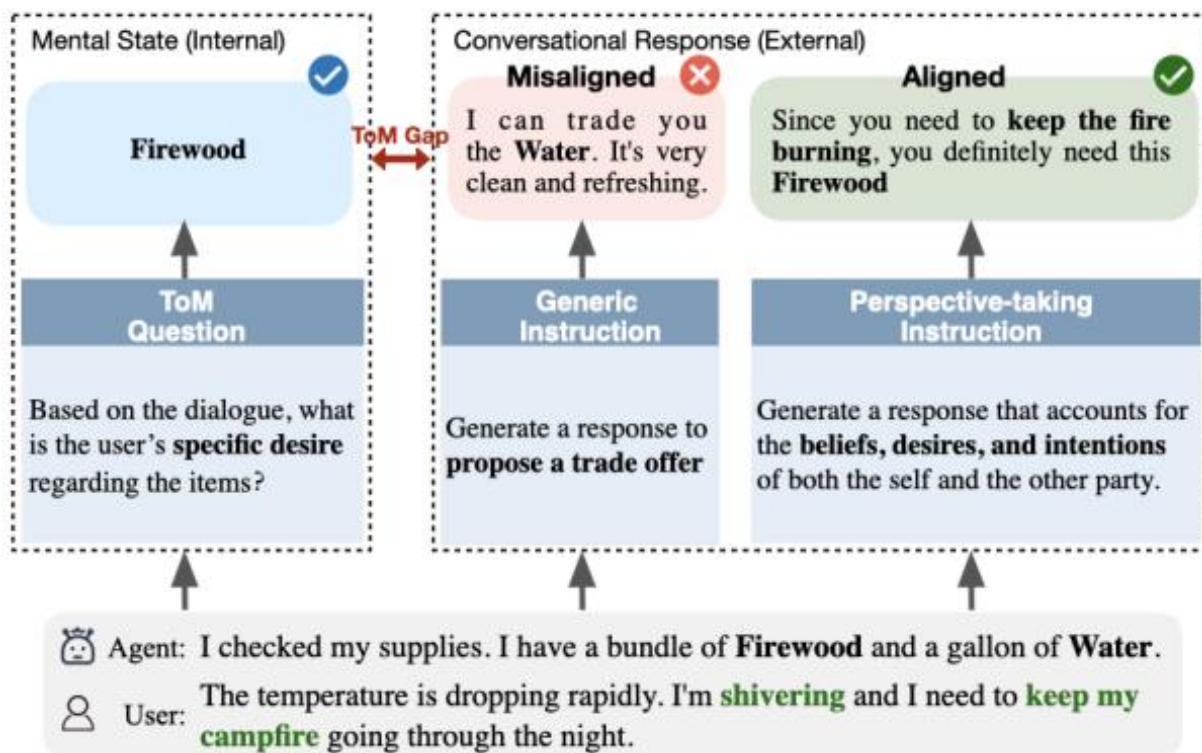


Figure 2: images/4fc664b1eccb8e7f00682e3bd887696c0cfa5972246e05e07fc4020cb5aeb625.jpg

Figure 1: A negotiation scenario illustrating the gap between ToM inference and ToM-aligned behavior in LLMs. (Left) The model correctly infers the user’s desire for firewood. (Middle) Under a generic task instruction, the model fails to apply this inferred mental state, producing an incoherent offer (i.e., water). (Right) When explicitly prompted to consider mental states, the model generates a contextually appropriate response.

图 1：一个谈判场景，展示了 LLMs 中 ToM 推理与 ToM 对齐行为之间的差距。（左）模型正确推断了用

户对柴火的渴望。(中)在通用任务指令下,模型未能应用这一推断出的心理状态,产生了一种不连贯的提议(即水)。(右)当明确提示模型考虑心理状态时,模型生成了一个符合情境的回应。

2024; Zhang et al., 2024) and recommendation (Li et al., 2026). With the rapid evolution of Large Language Models (LLMs) (Deng et al., 2023a, 2024), there is growing optimism that these models may have begun to exhibit ToM-like reasoning capabilities. Such claims have primarily been supported by recent benchmarks that probe LLMs' ability to interpret mental states under controlled structured scenarios (Jin et al., 2024; Shi et al., 2025; Wu et al., 2023; Zhang et al., 2025) or social contexts (Yu et al., 2025a; Chen et al., 2024b; Chan et al., 2024).

2024; Zhang 等人, 2024) 和推荐 (Li 等人, 2026)。随着大型语言模型 (LLMs) 的快速演进 (Deng 等人, 2023a, 2024), 人们越来越乐观地认为这些模型可能已经开始表现出类似心智理论 (ToM) 的推理能力。此类主张主要得到了最近基准测试的支持, 这些测试探究了 LLMs 在受控结构化场景下解读心理状态的能力 (Jin 等人, 2024; Shi 等人, 2025; Wu 等人, 2023; Zhang 等人, 2025) 或社会情境 (Yu 等人, 2025a; Chen 等人, 2024b; Chan 等人, 2024)。

However, a critical gap remains between these promising observations and the reliability of the underlying mechanisms. Although LLMs can infer human intentions to some extent, recent studies (Bortoletto et al., 2024; Ma et al., 2023; Jin et al., 2024; Shi et al., 2025) reveal that they fail to generalize to task-specific scenarios with genuine ToM reasoning. As illustrated in Figure 1 (Left vs. Middle), a critical misalignment exists between internal knowledge and external behavior:

然而, 这些有希望的观察结果与底层机制的可靠性之间仍存在关键差距。尽管大型语言模型 (LLMs) 在一定程度上可以推断人类意图, 但最近的研究 (Bortoletto 等, 2024; Ma 等, 2023; Jin 等, 2024; Shi 等, 2025) 表明, 它们无法在具有真实心智理论 (ToM) 推理的任务特定场景中进行泛化。如图 1 (左与中) 所示, 内部知识与外部行为之间存在关键的不匹配:

even when LLMs correctly answer ToM questions, their dialogue agents may still fail to negotiate effectively. Moreover, observed ToM-like behaviors often depend on carefully engineered prompts that scaffold perspective-taking (Li et al., 2023; Jung et al.; Sarangi et al., 2025; Chen et al.; Hou et al., 2024). As shown in Figure 1 (Middle vs. Right), once the explicit instruction to "infer and respond" replaced by a generic command, the model fails to ground its response in the mental states it implicitly encodes, reverting to incoherent generation. This suggests that current ToM-like behaviors may not reflect stable, intrinsic cognition, but instead ad hoc simulations triggered by instruction.

即使当大型语言模型 (LLMs) 正确回答心理理论 (ToM) 问题时, 它们的对话代理仍可能在协商方面表现不佳。此外, 观察到的类似心理理论的行为通常依赖于精心设计的提示, 这些提示引导着换位思考 (Li 等, 2023; Jung 等; Sarangi 等, 2025; Chen 等; Hou 等, 2024)。如图 1 (中间与右侧) 所示, 一旦将明确的“推断并回应”的指令替换为通用命令, 模型就无法将其回应建立在它隐含编码的心理状态之上, 而会退化为不连贯的生成。这表明, 当前的类似心理理论行为可能并不反映稳定的、内在的认知, 而是由指令触发的临时模拟。

Inspired by recent advances in mechanistic interpretability (Pan et al., 2024; Aljaafari et al., 2025; Yang et al., 2023; Chen et al., 2024a; Huben et al., 2024), we move beyond black-box prompt engineering and surface-level behavioral observation. We aim to uncover the intrinsic nature of social reasoning in LLMs, specifically investigating whether LLMs possess ToM-grounded social reasoning, how they are internally represented, and whether this internal knowledge can be effectively translated into stable, high-quality behaviors. Our investigation proceeds in three stages.

受近期在机制可解释性方面的进展启发 (Pan 等, 2024; Aljaafari 等, 2025; Yang 等, 2023; Chen 等, 2024a; Huben 等, 2024), 我们超越了黑盒提示工程和表面行为观察。我们的目标是揭示大语言模型 (LLMs) 中社会推理的内在本质, 具体探讨 LLMs 是否具备基于心智理论 (ToM) 的社会推理, 它们是如何内部表示的, 以及这种内部知识是否可以有效地转化为稳定、高质量的行为。我们的研究分三个阶段进行。

First, we seek to interpret the ToM reasoning capability within LLMs. We analyze activation patterns using causal tracing to identify whether ToM-specific features exist and locate where

they reside within the model stack. This leads to our first research question: (RQ1) In which layers does ToM-related information emerge and persist?

首先，我们试图在大型语言模型（LLMs）中解释心智理论（ToM）推理能力。我们利用因果追踪分析激活模式，以确定是否存在与 ToM 相关的特征，并确定这些特征在模型堆栈中的位置。这引出了我们第一个研究问题：(RQ1) 与 ToM 相关的信息在哪些层中出现并持续存在？

Second, identifying where ToM features exist offers a foundation for intervention. We examine whether steering internal activations can modulate the model’s ToM reasoning capabilities, moving from observation to control: (RQ2) To what extent can internal representations be leveraged to steer and improve ToM reasoning?

第二，识别 ToM 特征存在的位置为干预提供了基础。我们探讨通过引导内部激活是否能够调节模型的 ToM 推理能力，从观察转向控制：(RQ2) 内部表征在多大程度上可以被用来引导和提升 ToM 推理？

Finally, improvements on ToM benchmarks do not necessarily translate to better ToM-aligned behavior in downstream tasks. As inferring mental states is fundamental to predicting socially appropriate continuations (Yang et al., 2024; Cheng et al., 2024), genuine ToM alignment of LLMs should exhibit enhanced conversational performance. We therefore examine the downstream impact directly: (RQ3) Can manipulating these internal representations of LLMs effectively enhance response quality in dialogue tasks?

最后，对 ToM 基准的改进并不一定意味着在下游任务中能实现更好的 ToM 对齐行为。由于推断心理状态是预测社会适当延续的基础（Yang 等，2024；Cheng 等，2024），LLM 的真正 ToM 对齐应表现出增强的对话表现。因此，我们直接考察其下游影响：(RQ3) 通过操控 LLM 的这些内部表征，是否能有效提升对话任务中的响应质量？

To address these research questions, we introduce a novel and comprehensive framework for Causal-oriented Steering of ToM alignment in LLMs, named CoSToM. This framework aims to intrinsically align LLMs with ToM-like social reasoning by moving from interpretation to intervention. Specifically, CoSToM operates in two stages: it first identifies ToM-sensitive layers through causal tracing, and then steers these layers using activation manipulation. Given the dialogue history as input, causal tracing interprets the context encoder’s activations by probing them with ToM-focused questions, while activation steering supervises and adjusts these activations to better align the model’s internal representations with ToM-related features.

为了解决这些问题，我们引入了一个新颖且全面的框架，用于面向因果的共情对齐（ToM alignment）在大型语言模型（LLMs）中的引导，该框架命名为 CoSToM。该框架旨在通过从解释转向干预，使 LLMs 内在地与类似共情的社会推理（ToM-like social reasoning）对齐。具体而言，CoSToM 分为两个阶段：首先，通过因果追踪识别出对共情敏感的层；然后，使用激活操控引导这些层。给定对话历史作为输入，因果追踪通过使用聚焦于共情的问题来解释上下文编码器的激活情况；而激活引导则监督并调整这些激活，以更好地使模型内部表示与共情相关特征对齐。

Our contributions are as follows:

我们的贡献如下：

- ToM Interpretation: We systematically trace ToM-related features across layer-wise activations in LLMs, revealing that these features are predominantly encoded in early layers of LLMs.
- Efficient and Lightweight ToM Intervention: We propose CoSToM, a lightweight alignment framework that induces stable, human-like social reasoning via targeted activation steering, requiring updates to only a small subset of parameters in the identified ToM-critical layers.
- Enhancement on Dialogue Tasks: Experiments on negotiation and persuasion dialogues

demonstrate that internal ToM alignment via CoSToM leads to substantial improvements in dialogue quality. Notably, CoSToM functions as a plug-and-play module that generalizes effectively across diverse social interaction tasks.²

- ToM 解释：我们系统地追踪 LLMs 中与 ToM 相关的特征在各层激活中的表现，揭示出这些特征主要编码在 LLMs 的早期层中。
- 高效且轻量的 ToM 干预：我们提出了 CoSToM，一个轻量级的对齐框架，通过有针对性的激活引导，诱导出稳定、类似人类的社会推理，仅需对识别出的 ToM 关键层中的一小部分参数进行更新。
- 对对话任务的增强：在谈判和说服对话上的实验表明，通过 CoSToM 实现的内部 ToM 对齐显著提升了对话质量。值得注意的是，CoSToM 作为一个即插即用模块，在各种不同的社会交互任务中表现出良好的泛化能力。²

2 Related Work

2 相关工作

ToM in LLMs Recent work on ToM in LLMs has focused on evaluating and enhancing their ability to infer mental states such as beliefs and intentions, often using benchmarks adapted from classical psychological tests (Shi et al., 2025; Jin et al., 2024; Xu et al., 2024). To address observed performance limitations, existing approaches primarily adopt either prompt-based scaffolding, which elicits ToM reasoning through carefully engineered instructions (Wilf et al., 2024; Jung et al.; Sarangi et al., 2025; Chen et al.; Hou et al., 2024; Sclar et al.), or neuro-symbolic and Bayesian frameworks

ToM 在 LLMs 中近期关于 ToM 在 LLMs 中的研究主要集中在评估和提升其推断诸如信念和意图等心理状态的能力，通常使用从经典心理测试中改编的基准测试 (Shi 等, 2025; Jin 等, 2024; Xu 等, 2024)。为了解决观察到的性能限制，现有方法主要采用两种方式：一种是基于提示的支架方法，通过精心设计的指令引发 ToM 推理 (Wilf 等, 2024; Jung 等; Sarangi 等, 2025; Chen 等; Hou 等, 2024; Sclar 等)，或者神经符号和贝叶斯框架

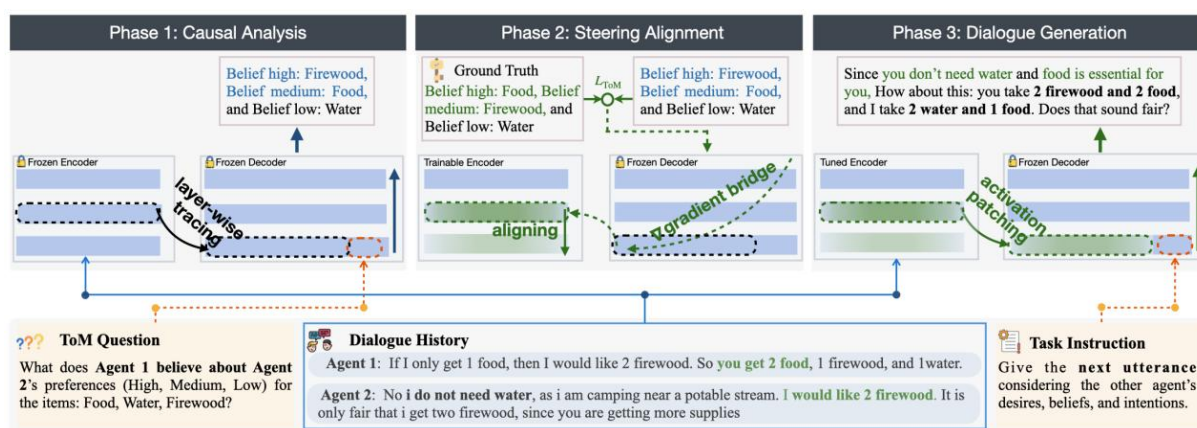


Figure 3: images/155a96e10a13ad11202b877409278b5b5645e1c2c1f768f6cda033a23d1a61f7.jpg

Figure 2: The overview of the CoSToM framework.

图 2: CoSToM 框架的概览。

that integrate LLMs with explicit cognitive models for mental-state inference (Chandra et al.,

2023; Miao et al., 2022; Baker et al., 2017; Jin et al., 2024; Shi et al., 2025; Zhang et al., 2025). While effective in structured settings, these methods rely on external scaffolding and offer limited insight into or control over ToM representations inside LLMs.

将大型语言模型 (LLMs) 与显式认知模型结合, 以进行心理状态推理 (Chandra 等, 2023; Miao 等, 2022; Baker 等, 2017; Jin 等, 2024; Shi 等, 2025; Zhang 等, 2025)。尽管这些方法在结构化环境中是有效的, 但它们依赖于外部支架, 并且对大型语言模型内部的理论-of-mind (ToM) 表示提供有限的洞察或控制。

ToM in Dialogue Agents Beyond static benchmarks (e.g., Sally-Anne tests), recent work has increasingly evaluated ToM within dynamic dialogue settings, where agents must maintain contextually appropriate and socially sensitive interactions, such as persuasion (Yu et al., 2025b), negotiation (Chan et al., 2024), education (Saha et al., 2023), stress testing (Kim et al., 2023), and recommendation (Li et al., 2026). To improve performance in these settings, several approaches aim to align dialogue responses with inferred internal mental states (Sicilia and Alikhani, 2025; Jafari et al., 2025; Qiu et al., 2024). For example, MindDial (Qiu et al., 2024) explicitly tracks beliefs to guide response generation, while Jafari et al. (2025) enforce logical consistency by refining ToM-related decoders. However, bridging the gap between ToM reasoning and dialogue generation remain challenging. Appending inferred mental states as text can propagate errors, while fine-tuning on ToM QA tasks often fails to translate improved reasoning into socially aligned dialogue due to task misalignment. In contrast, CoSToM bypassed the uncertainty, directly transplanting the causal reasoning activations into the decoder to drive the dialogue generation.

在对话代理中, 除了静态基准 (例如, Sally-Anne 测试) 之外, 近期的研究越来越多地在动态对话环境中评估心智理论 (ToM), 其中代理必须维持情境适当且社会敏感的互动, 例如说服 (Yu 等, 2025b)、谈判 (Chan 等, 2024)、教育 (Saha 等, 2023)、压力测试 (Kim 等, 2023) 和推荐 (Li 等, 2026)。为了在这些环境中提高性能, 一些方法旨在将对话响应与推断出的内部心理状态对齐 (Sicilia 和 Alikhani, 2025; Jafari 等, 2025; Qiu 等, 2024)。例如, MindDial (Qiu 等, 2024) 明确追踪信念以指导响应生成, 而 Jafari 等 (2025) 通过改进与 ToM 相关的解码器来强制逻辑一致性。然而, 弥合 ToM 推理与对话生成之间的差距仍然具有挑战性。将推断出的心理状态作为文本附加可能会传播错误, 而在 ToM QA 任务上进行微调通常由于任务错配而无法将改进的推理转化为符合社会规范的对话。相比之下, CoSToM 略过了不确定性, 直接将因果推理激活量移植到解码器中以驱动对话生成。

Mechanistic Interpretability in LLMs Mechanistic interpretability (Zhao et al., 2024a; Singh et al., 2024) seeks to reverse engineer the “black box” of neural networks by uncovering how high-

LLM 中的机制可解释性机制可解释性 (Zhao 等, 2024a; Singh 等, 2024) 旨在通过揭示高-

level concepts are encoded in latent representations (Zhao et al., 2024c,b, 2025; Deng et al., 2025; Azaria and Mitchell, 2023). Recent work has further developed causal and intervention-based tools for analyzing internal model states, including causal mediation analysis for explaining model behavior (Stolfo et al., 2023; Cheng et al., 2022; Tian et al., 2024; Cheang et al., 2025) and activation-level methods for locating or manipulating encoded information, such as activation patching, linear representation analysis, and factual model editing (Dumas et al., 2025; Tigges et al., 2024; Meng et al., 2022). Despite these advances, mechanistic analyses of machine ToM remain underexplored, and even existing studies are primarily diagnostic, focusing on where information is encoded or how it contributes to model predictions. By contrast, our work applies causal tracing not only to interpret ToM-relevant representations, but also to steer them, enabling direct and effective improvement of ToM reasoning.

层级概念被编码在潜在表示中 (Zhao 等, 2024c,b, 2025; Deng 等, 2025; Azaria 和 Mitchell, 2023)。最近的研究进一步开发了用于分析内部模型状态的因果和干预工具, 包括因果中介分析, 用于解释模型行为 (Stolfo 等, 2023; Cheng 等, 2022; Tian 等, 2024; Cheang 等, 2025), 以及用于定位或操控编码信息的激活层级方法, 例如激活补丁、线性表示分析和事实模型编辑 (Dumas 等, 2025; Tigges 等, 2024; Meng 等, 2022)。尽管取得了这些进展, 机器 ToM 的机制分析仍处于初步阶段, 甚至现有的研究也主要是诊断性的, 关注信息是如何被编码或如何影响模型预测的。相比之下, 我们的工作不仅将因果追踪应用于解释与 ToM 相关的表示, 还用于引导这些表示, 从而实现了对 ToM 推理的直接且有效的改进。

3 Methodology

3 方法论

The overview of the COSTOM framework is illustrated in Figure 2. This section is structured around our research questions, with each subsection detailing the corresponding methodological approach.

COSTOM 框架的概述如图 2 所示。本节围绕我们的研究问题展开，每个子部分详细介绍了相应的研究方法。

3.1 Interpreting ToM: Locating ToM Representations via Causal Tracing

3.1 解释 ToM：通过因果追踪定位 ToM 表征

The first phase of CoSToM aims to identify whether and where ToM capabilities are instantiated within the model. We hypothesize that if an LLM genuinely understands a social scenario, the mental states of the agent, specifically belief, desire, and intention (BDI), must be encoded in its internal activations. To verify this, we employ

CoSToM 的第一阶段旨在识别 ToM 能力是否以及在模型的何处被实现。我们假设，如果一个大语言模型真正理解了一个社交场景，那么代理的内心状态，特别是信念、欲望和意图（BDI），必须在其内部激活中被编码。为了验证这一点，我们采用

the causal tracing to “read” these implicit mental states from the model.

对模型进行因果追溯，以“读取”这些隐含的心理状态。

Given a dialogue history x and a target LLM with multiple layers, we extract the hidden activation at a specific layer ℓ while the model processes x , denoted as $h^\ell(x)$. Intuitively, if $h^\ell(x)$ contains ToM-related information, it should be possible to decode the corresponding mental state directly from the activation. Operationally, we instantiate two copies of the same LLM: a context encoder and a probe decoder. The encoder processes the dialogue history and produces intermediate activations. We then inject the frozen activation $h_{\text{enc}}^\ell(x)$ from layer ℓ of the encoder into the decoder, which is tasked with answering a ToM-focused question q (e.g., inferring an agent’s belief). The decoder’s output is given by

给定一个对话历史 x 和一个具有多个层的目标大语言模型（LLM），我们在模型处理 x 时提取特定层 ℓ 的隐藏激活状态，记作 $h^\ell(x)$ 。直觉上，如果 $h^\ell(x)$ 包含心智理论（ToM）相关的信息，那么应该可以直接从激活状态中解码出相应的心理状态。操作上，我们实例化两个相同 LLM 的副本：一个上下文编码器和一个探针解码器。编码器处理对话历史并生成中间激活状态。然后，我们将编码器中层 ℓ 的冻结激活状态 $h_{\text{enc}}^\ell(x)$ 注入到解码器中，解码器的任务是回答一个以心智理论为中心的问题 q （例如，推断代理的信念）。解码器的输出为

$$\tilde{y}_\ell = f_{\text{dec}}(q \mid h_{\text{enc}}^\ell(x)).$$

By evaluating the decoder’s accuracy in answering ToM-related questions based solely on these patched activations, we empirically determine which layers contain the necessary information to reconstruct the agents’ mental states.

通过仅基于这些修补后的激活值评估解码器在回答与心智理论（ToM）相关问题时的准确性，我们实证地确定了哪些层包含重构智能体心理状态所需的信息。

3.2 Steering ToM: Aligning Mental States via Activation Intervention

3.2 指导 ToM：通过激活干预对心理状态进行对齐

Building upon the identification of ToM-sensitive layers, we next examine whether directly steering internal activations can modulate ToM reasoning capabilities. To this end, we move beyond passive interpretation toward active alignment. While causal tracing reveals where the information resides, steering alignment focuses on how to refine these representations. Our core intuition is to leverage the frozen probe decoder as a differentiable verifier to steer the context encoder’s latent representations towards accurate social reasoning.

在识别出与心智理论（ToM）敏感的层之后，我们接下来探讨是否可以通过直接引导内部激活来调节心智理论推理能力。为此，我们超越被动解释，转向主动对齐。虽然因果追踪揭示了信息所在的位置，但引导对齐则关注如何优化这些表示。我们的核心直觉是利用冻结的探针解码器作为可微分的验证器，引导上下文编码器的潜在表示朝着准确的社会推理方向发展。

Steering Objective We formulate a supervised steering objective that explicitly aligns internal activations with ground-truth mental states. Specifically, we employ the same dual-model setup, where the decoder receives the patched activations from the encoder and is prompted with specific ToM questions (e.g., For each agent, what are their desires (High, Medium, Low) for the items: food, water, and firewood?). By comparing the probability distribution generated by the decoder against the ground-truth BDI labels y' , we calculate a standard cross-entropy loss:

转向目标我们提出一个监督式的转向目标，明确地将内部激活与真实的心理状态对齐。具体来说，我们采用相同的双模型设置，其中解码器接收来自编码器的补丁激活，并被提示特定的 ToM 问题（例如，对于每个代理，他们对物品（食物、水和柴火）的欲望（高、中、低）是什么？）。通过将解码器生成的概率分布与真实 BDI 标签 y' 进行比较，我们计算出一个标准的交叉熵损失：

$$\mathcal{L}_{\text{ToM}} = -\log P_{\text{dec}}(y' \mid h_{\text{enc}}^{\ell}(x), q).$$

Gradient Bridge Mechanism We backpropagate the calculated loss L_{ToM} through the network. Distinct from standard fine-tuning, CoSToM establishes a gradient bridge via the activation space. Crucially, although the decoder is kept frozen, it functions as a transparent conduit: gradients derived from the output loss traverse backwards through the decoder, cross the patched activation interface, and flow upstream into the context encoder. Since the activations are intercepted at a specific layer ℓ , the gradients propagate backwards only through the layers preceding this interface (Layers 0 to ℓ). Consequently, only the LoRA adapters installed in these shallow layers are updated, while the deeper layers of the encoder remain frozen and computationally uninvolved. By doing so, we effectively “steer” the encoder to spontaneously generate ToM-enriched representations with minimal parameter updates.

梯度桥机制我们通过网络反向传播计算出的损失 L_{ToM} 。与标准微调不同，CoSToM 通过激活空间建立了一个梯度桥。关键的是，尽管解码器被保持冻结状态，它却作为一个透明的通道：从输出损失导出的梯度会通过解码器反向传播，穿过修补的激活接口，并向上流至上下文编码器。由于激活在特定层 ℓ 被截获，梯度仅通过该接口之前的层（层 0 到 ℓ ）反向传播。因此，只有安装在这些浅层的 LoRA 适配器被更新，而编码器的更深层则保持冻结并未参与计算。通过这样做，我们有效地“引导”编码器以最小的参数更新自发生成富含 ToM 的表示。

Efficiency and Scalability Although the dual-model architecture requires simultaneous loading of the context encoder and the probe decoder, the memory footprint remains linear ($2N$) relative to the base model size. Furthermore, since CoSToM utilizes Parameter-Efficient Fine-Tuning (PEFT) to update only a sparse set of LoRA adapters in the identified ToM-critical layers, the number of trainable parameters is significantly lower than that of full-layer fine-tuning. And this architecture is inherently compatible with standard distributed training strategies (e.g., FSDP or ZeRO-3), allowing the $2N$ footprint to be sharded across GPU nodes. This ensures that our framework can

be seamlessly extended to large-scale models without encountering theoretical or engineering bottlenecks.

效率和可扩展性尽管双模型架构需要同时加载上下文编码器和探针解码器，但其内存占用量仍与基础模型大小呈线性关系（ $2N$ ）。此外，由于 CoSToM 使用参数高效微调（PEFT）技术，仅在识别出的 ToM 关键层中更新一组稀疏的 LoRA 适配器，因此可训练参数的数量显著少于全层微调。并且，这种架构本质上与标准的分布式训练策略（例如 FSDP 或 ZeRO-3）兼容，允许将 $2N$ 的内存占用量分布在多个 GPU 节点上。这确保了我们的框架可以无缝扩展到大规模模型，而不会遇到理论或工程上的瓶颈。

3.3 Leveraging ToM: Enhancing Downstream Dialogue Generation

3.3 利用心智理论：提升下游对话生成

Achieving high accuracy on static ToM benchmarks does not necessarily translate into ToM-aligned behavior in interactive settings. Therefore, the final phase of CoSToM focuses on validating whether these aligned internal representations can effectively translate from internal reasoning to external action. During inference, we deploy the ToM-enriched context encoder tuned in Section 3.2. In contrast to training, where the decoder serves as a verifier for mental-state inference, it now assumes its standard role as a generator. The inference pipeline operates as follows:

在静态 ToM 基准测试中取得高准确性并不一定意味着在交互式环境中表现出与 ToM 对齐的行为。因此，CoSToM 的最后阶段专注于验证这些对齐的内部表示是否能够有效地将内部推理转化为外部行为。在推理过程中，我们部署了在第 3.2 节中调优的 ToM 增强型上下文编码器。与训练过程中解码器作为心理状态推理的验证者不同，现在它重新承担其标准的生成器角色。推理流程如下：

1) Encoding: Given a dialogue history x , the tuned

1) 编码：给定一个对话历史 x ，经过调整的

encoder produces latent representations $h_{\text{enc}}^{\ell'}(x)$ at the ToM-sensitive layers identified in Section 3.1. Importantly, these representations implicitly encode accurate beliefs, desires, and intentions.

编码器在第 3.1 节中确定的 ToM 敏感层产生潜在表示 $h_{\text{enc}}^{\ell'}(x)$ 。重要的是，这些表示隐式地编码了准确的信念、欲望和意图。

2) Generation: The ToM-enriched activations are then provided to the frozen decoder, which is prompted with task-specific instructions q_{task} (e.g., negotiation or persuasion objectives), instead of ToM-focused questions used in previous phases. The response r is generated as

2) 生成：随后，这些富含心智理论（ToM）的激活值被提供给冻结的解码器，该解码器被提示以任务特定的指令 q_{task} （例如协商或说服目标），而不是之前阶段中使用的心智理论相关问题。响应 r 生成如下：

$$r = f_{\text{dec}}(q_{\text{task}} \mid h_{\text{enc}}^{\ell'}(x)),$$

where conditioning on the refined ToM-related internal states enables the decoder to translate the encoder’s internal ToM reasoning into coherent and socially appropriate dialogue actions.

当对精炼的 ToM 相关内部状态进行条件建模时，解码器能够将编码器的内部 ToM 推理转化为连贯且符合社会规范的对话行为。

4 Experiments

4 实验

4.1 Experimental Setups

4.1 实验设置

Dataset. We adopt NEGOTIATIONTom (Chan et al., 2024) and PERSUASIVETOM (Yu et al., 2025b) for evaluation. Detailed statistics are presented in Table 3. To assess downstream dialogue quality across diverse conversational phases, we curate stratified test subset ($N = 100$ for NEGOTIATIONTom, $N = 200$ for PERSUASIVETOM). These subsets are randomly sampled from the beginning, middle, and final stages of the interaction with a fixed ratio of $1 : 2 : 1$.

数据集. 我们采用 NEGOTIATIONTom (Chan 等, 2024) 和 PERSUASIVETOM (Yu 等, 2025b) 进行评估。详细统计结果如表 3 所示。为了评估对话质量在不同对话阶段的表现, 我们整理了分层测试子集 ($N = 100$ 用于 NEGOTIATIONTom, $N = 200$ 用于 PERSUASIVETOM)。这些子集是从交互的开始、中间和结束阶段随机抽取的, 固定比例为 $1 : 2 : 1$ 。

Baselines. We evaluate CoSToM against five representative baselines to ensure a comprehensive comparison across different paradigms in the dialogue generation task:

基线. 我们针对五种具有代表性的基线进行评估, 以确保在对话生成任务中跨不同范式的全面比较:

(1) Prompting-based Baselines: (i) Zero-shot: Directly prompt LLM to generate the next utterance without task-specific training. (ii) MindDial (Qiu et al., 2024): An explicit reasoning method that first infers the partner’s BDI states and then generates a response conditioned on these ToM estimations.

(2) Finetuning-based Baselines: To ensure a fair comparison, all tuning-based baselines are trained on the same dataset as CoSToM. (iii) MindDial (Fine-tuned) (Qiu et al., 2024): A supervised fine-tuned version of MindDial optimized on the same ToM-centric QA pairs to align its explicit reasoning. (iv) Full-Layer LoRA: A standard parameter-efficient fine-tuning baselines in which LoRA adapters are applied to the same ToM-sensitive encoder layers as in CoSToM, as well as all layers of the decoder. All these adapters are tuned for the ToM tasks. (v) LatentQA (Pan et al.,

(1) 基于提示的基线方法: (i) 零样本: 直接提示 LLM 生成下一个对话内容, 而无需进行特定任务的训练。(ii) MindDial (Qiu 等, 2024): 一种显式推理方法, 首先推断对话伙伴的 BDI 状态, 然后根据这些心智理论 (ToM) 估计生成响应。

(2) 基于微调的基线方法: 为确保公平比较, 所有基于微调的基线方法均在与 CoSToM 相同的语料库上进行训练。(iii) MindDial (微调版) (Qiu 等, 2024): 一种监督微调版本的 MindDial, 其在相同的以心智理论为中心的问答对上进行了优化, 以对齐其显式推理。(iv) 全层 LoRA: 一种标准的参数高效微调基线方法, 其中 LoRA 适配器被应用于与 CoSToM 相同的 ToM 敏感编码器层, 以及解码器的所有层。所有这些适配器均针对 ToM 任务进行微调。(v) LatentQA (Pan 等,

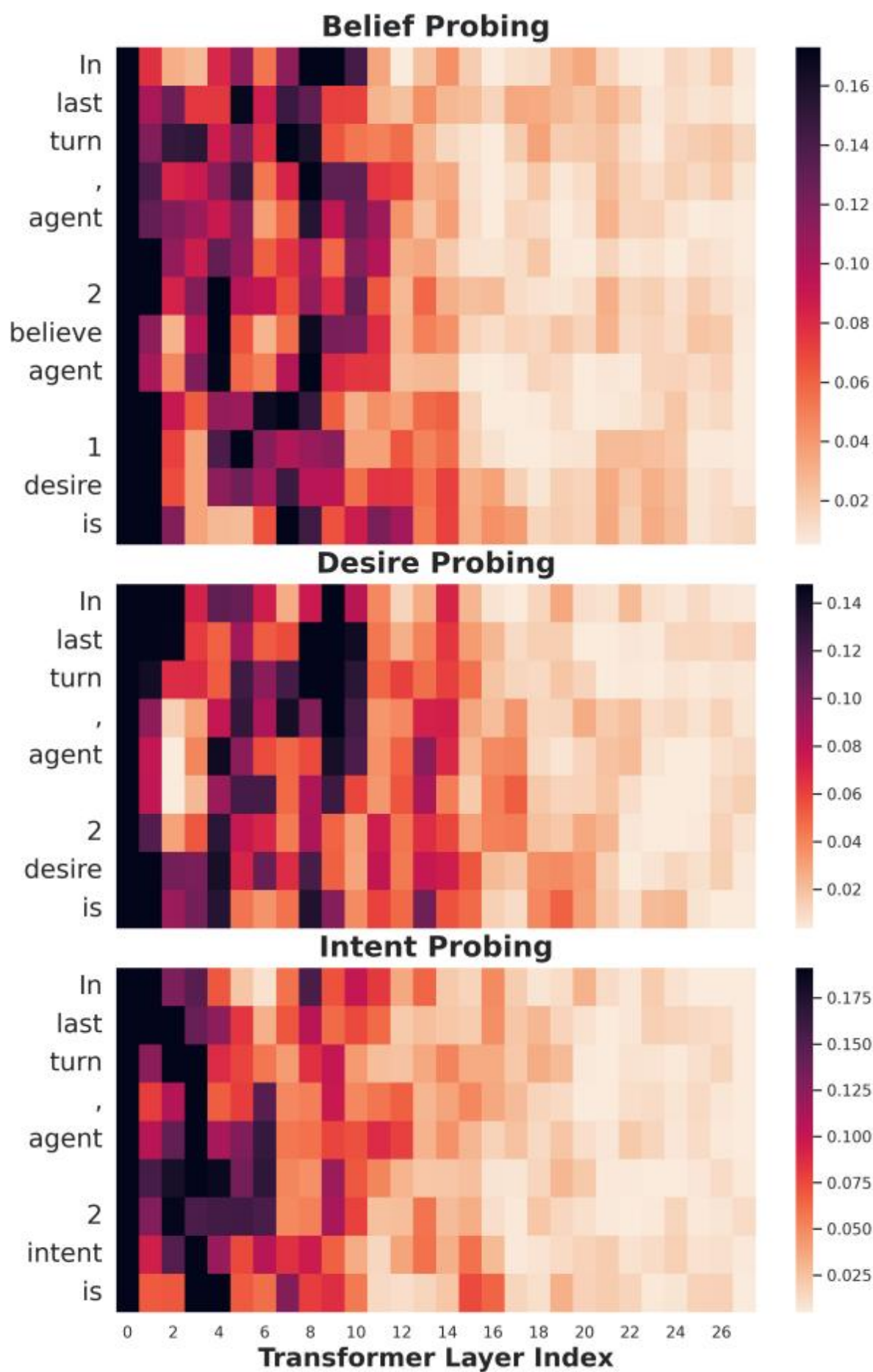


Figure 3: Layer-wise probing results on NEGOTIATION-TOM with Qwen2.5. Details and corresponding results of Llama-3 are provided in Appendix B.

图 3：在 NEGOTIATION-TOM 上使用 Qwen2.5 的逐层探查结果。Llama-3 的详细信息及对应结果见附录 B。

2024): A dual-model architecture proposed by Jafari et al. (2025) that decodes latent ToM signals by fine-tuning the decoder. Detailed instruction prompts and implementation details are provided in Appendices E and A, respectively.

2024): Jafari 等人（2025）提出的一种双模型架构，通过微调解码器来解码潜在的 ToM 信号。详细的指令提示和实现细节分别在附录 E 和 A 中提供。

4.2 RQ1: ToM Interpretation

4.2 RQ1: 心理理论解释

To answer RQ1 (Where does ToM-related information emerge and persist?), we analyze the reconstruction performance of mental states (belief, desire, and intention) across different layers of the context encoder. Table 1 and 2 present the quantitative results for the NEGOTIATION TOM and PERSUASIVETOM tasks, respectively. Our causal tracing experiments yield three critical observations:

为回答 RQ1 (心智理论相关信息在何处产生并持续存在?), 我们分析了在上下文编码器的不同层次上对心理状态 (信念、欲望和意图) 的重建性能。表 1 和表 2 分别展示了 NEGOTIATION TOM 和 PERSUASIVETOM 任务的定量结果。我们的因果追踪实验得出三个关键观察结果：

1) The “Early Layer Primacy” of ToM encoding. Contrary to the conventional assumption that high-level reasoning resides solely in deeper layers (Song et al., 2025; Yang et al., 2025), our results reveal that ToM representations are predominantly localized within the model’s shallow layers. As shown in Table 1 and 2, both Llama-3 and Qwen2.5 exhibit a distinct “ToM-sensitive zone” within the

1) ToM 编码的“早期层优先性”。与传统假设（Song 等，2025；Yang 等，2025）不同，认为高级推理仅存在于模型的深层结构中，我们的结果表明，ToM 表示主要集中在模型的浅层结构中。如表 1 和表 2 所示，Llama-3 和 Qwen2.5 都在其内部表现出一个明显的“ToM 敏感区域”。

Layer	Llama-3-8B-Instruct						Qwen2.5-7B-Instruct					
	Intent		Desire		Belief		Intent		Desire		Belief	
Agent	Agent	Agent	Agent	Agent	Agent	Agent	Agent	Agent	Agent	Agent	Agent	Agent
1	2	1	2	1	2	1	2	1	2	1	2	
Base	18.00	11.81	39.80	44.59	19.55	27.43	8.30	8.58	37.83	38.82	20.82	23.21
0	12.66	14.35	37.41	34.60	24.05	20.53	15.05	21.10	35.44	36.71	25.60	23.07
2	13.92	13.64	40.79	37.41	25.04	27.85	15.19	19.83	34.88	37.27	28.13	25.32
3	15.05	16.32	40.08	34.60	23.91	25.88	13.92	18.42	36.29	37.41	26.30	22.36
4	13.50	15.19	40.23	34.60	24.05	22.22	7.17	13.78	35.86	32.91	24.61	23.49
6	9.28	13.92	38.96	35.44	23.21	25.74	6.89	13.92	33.33	33.47	24.19	24.47
8	8.30	9.85	33.47	26.30	16.74	15.19	5.20	9.99	30.28	28.27	26.02	25.60
10	5.77	6.75	22.36	15.61	11.53	11.95	5.06	7.17	24.75	26.72	27.14	24.33
15	7.31	4.22	8.44	7.17	9.56	5.20	10.97	8.16	24.75	21.10	21.52	22.22

20	3.94	3.94	5.06	8.30	11.11	5.49	6.61	8.02	21.52	22.08	21.10	21.80
24	4.78	1.27	2.39	7.88	8.86	5.91	7.03	7.31	12.66	13.08	5.77	6.61

层	Llama-3-8B-Instruct						Qwen2.5-7B-Instruct					
意图	欲望			信念			意图	欲望				
代理 1	Agent 2	代理 1	Agent 2	代理 1	Agent 2	代理 1	代理 2	代理 1	代理 2	代理 1	代理 2	
基础	18.00	11.81	39.80	44.59	19.55	27.43	8.30	8.58	37.43	8.58	37.43	
0	12.66	14.35	37.41	34.60	24.05	20.53	15.05	21.10	35.43	21.10	35.43	
2	13.92	13.64	40.79	37.41	25.04	27.85	15.19	19.83	34.60	19.83	34.60	
3	15.05	16.32	40.08	34.60	23.91	25.88	13.92	18.42	36.04	18.42	36.04	
4	13.50	15.19	40.23	34.60	24.05	22.22	7.17	13.78	35.43	13.78	35.43	
6	9.28	13.92	38.96	35.44	23.21	25.74	6.89	13.92	33.96	13.92	33.96	
8	8.30	9.85	33.47	26.30	16.74	15.19	5.20	9.99	30.79	9.99	30.79	
10	5.77	6.75	22.36	15.61	11.53	11.95	5.06	7.17	24.05	7.17	24.05	
15	7.31	4.22	8.44	7.17	9.56	5.20	10.97	8.16	24.05	8.16	24.05	
20	3.94	3.94	5.06	8.30	11.11	5.49	6.61	8.02	21.52	8.02	21.52	
24	4.78	1.27	2.39	7.88	8.86	5.91	7.03	7.31	12.66	7.31	12.66	

Table 1: Causal Tracing results on the NEGOTIATIONTOM dataset. The table compares reconstruction accuracy across layers for both Llama-3 and Qwen2.5. Bold indicates the best performance for each metric. Results confirm that ToM information is predominantly encoded in the shallow layers (e.g., Layer (0 - 3) for both models.

表 1: 在 NEGOTIATIONTOM 数据集上的因果追溯结果。该表比较了 Llama-3 和 Qwen2.5 在不同层上的重建精度。加粗表示每个指标的最佳性能。结果确认了 ToM 信息主要编码在浅层（例如，两个模型的第 0-3 层）。

Layer	Llama-3-8B-Instruct						Qwen2.5-7B-Instruct					
Intent	Desire			Belief			Intent	Desire			Belief	
Persuader	Persuader	Persuader	Persuader	Persuader	Persuader	Persuader	Persuader	Persuader	Persuader	Persuader	Persuadee	Persuadee
Base	40.35	87.78	37.88	68.93	65.85	62.06	40.35	90.75	98.45	70.29	77.50	82.26
0	41.90	88.48	42.26	71.66	60.16	58.13	41.13	91.08	93.55	72.47	68.56	80.78
2	42.43	87.13	51.03	70.03	60.98	49.51	42.42	91.42	95.36	70.29	67.75	78.57
4	39.33	86.47	52.84	67.30	59.62	48.77	43.95	89.77	79.64	66.21	65.31	66.75
6	43.44	88.12	52.06	65.94	55.83	47.54	42.41	91.08	86.34	61.58	71.27	69.70
10	20.82	72.61	49.74	59.13	42.55	35.96	44.47	90.09	58.50	61.58	60.43	50.49
15	12.34	34.65	31.44	24.80	20.33	18.97	43.70	88.44	64.17	47.68	43.36	37.43
20	8.22	22.77	17.78	10.63	9.21	11.82	31.36	67.88	57.47	40.59	39.29	41.62
24	4.63	8.58	7.99	7.09	6.78	5.91	35.21	79.53	69.07	38.14	42.54	41.87
27	5.14	5.94	4.12	1.08	2.98	3.20	36.24	76.23	43.81	28.61	25.20	27.09

层	Llama-3-8B-Instruct					Qwen2.5-7B-Instruct			
意图		欲望		信念		意图		欲望	
说服者	说服对象	说服者	说服对象	说服者	说服对象	说服者	说服对象	说服者	说服对象
基础	40.35	87.78	37.88	68.93	65.85	62.06	40.35	90.75	
0	41.90	88.48	42.26	71.66	60.16	58.13	41.13	91.08	
2	42.43	87.13	51.03	70.03	60.98	49.51	42.42	91.42	
4	39.33	86.47	52.84	67.30	59.62	48.77	43.95	89.77	
6	43.44	88.12	52.06	65.94	55.83	47.54	42.41	91.08	
10	20.82	72.61	49.74	59.13	42.55	35.96	44.47	90.09	
15	12.34	34.65	31.44	24.80	20.33	18.97	43.70	88.44	
20	8.22	22.77	17.78	10.63	9.21	11.82	31.36	67.88	
24	4.63	8.58	7.99	7.09	6.78	5.91	35.21	79.53	
27	5.14	5.94	4.12	1.08	2.98	3.20	36.24	76.23	

Table 2: Causal Tracing results on the PERSUASIVETOM dataset. The table compares mental state reconstruction accuracy across layers for both Llama-3 and Qwen2.5. Bold highlights the peak performance among probed layers. Similar to NEGOTIATIONTOM, critical ToM information is concentrated in the shallow-to-middle layers.

表 2：在 PERSUASIVETOM 数据集上的因果追溯结果。该表比较了 Llama-3 和 Qwen2.5 在不同层上的心理状态重建准确性。粗体突出显示了被探测层中的峰值性能。与 NEGOTIATIONTOM 类似，关键的 ToM 信息集中在浅层到中层。

Dataset	train	val	eval
NEGOTIATIONTOM	1,335	334	711
PERSUASIVETOM	10,355	2,219	2,222

数据集	火车	val	eval
谈判 TOM	1,335	334	711
有说服力的沟通	10,355	2,219	2,222

Table 3: Statistics of the NEGOTIATIONTom and PERSUASIVETom datasets used in our experiments.

表 3：我们在实验中使用的 NEGOTIATIONTom 和 PERSUASIVETom 数据集的统计信息。

initial stages (e.g., $L_0 - L_3$). For instance, in the negotiation task, the decodability of desire peaks at Layer 2 ($\sim 37\%$) and remains high through Layer 6, suggesting that fundamental social information is extracted almost following the embedding projection. This observation is further validated by layer-wise probing (Figure 3), where classification accuracy, as a proxy for knowledge density, shows a high concentration of ToM-specific information in the early layers.

初始阶段（例如， $L_0 - L_3$ ）。例如，在谈判任务中，欲望的可解码性在第 2 层（ $\sim 37\%$ ）达到峰值，并在第 6 层保持较高水平，这表明基本的社会信息几乎是在嵌入投影之后立即被提取的。这一观察结果通过逐层探针（图 3）得到了进一步验证，在图中，分类准确率作为知识密度的代理指标，显示出早期层中存在大量心智理论（ToM）特定的信息。

2) Representational depth of mental state. Causal tracing results reveal that the decodability of mental states is intrinsically tied to their semantic complexity: intention consistently proves as more complicated dimension, yielding significantly lower accuracy compared to belief or desire (15% vs. 40% in negotiation). Notably, Llama-3

2) 心态状态的表征深度。因果追踪结果揭示，心态状态的可解码性与其语义复杂性内在相关：意图始终证明为更复杂的维度，其准确率显著低于信念或欲望（谈判中为 15% vs. 40%）。值得注意的是，Llama-3

(persuader role) exhibits a “staged maturation” of these states that mirrors human cognitive progress (Rao and Georgeff, 1991; Wooldridge, 2003; Wellman and Liu, 2004; Bratman, 1987): representational peaks shift from Layer 2 for belief to Layer 4 for desire, and finally to Layer 6 for intention.

(说服者角色) 表现出这些状态的“分阶段成熟”，这与人类认知进步相呼应（Rao 和 Georgeff, 1991; Wooldridge, 2003; Wellman 和 Liu, 2004; Bratman, 1987）：表征高峰从用于信念的第 2 层转移到用于欲望的第 4 层，最后转移到用于意图的第 6 层。

4.3 RQ2: Efficacy of CoSToM

4.3 RQ2: CoSToM 的效果

We evaluate the impact of causal-oriented steering on the mental state alignment. Quantitative results for Llama-3 are presented in Figures 4 and 5, with parallel results for Qwen2.5 are provided in Appendix C. The experimental evidence highlights two primary advantages of CoSToM: Stability and Magnitude.

我们评估了因果导向引导对心理状态对齐的影响。Llama-3 的定量结果在图 4 和图 5 中呈现，Qwen2.5 的平行结果在附录 C 中提供。实验结果突出了 CoSToM 的两个主要优势：稳定性与幅度。

1) Stability: mitigating representation collapse. A striking observation is that CoSToM effectively counteracts the “vanishing ToM” phenomenon. In baseline model (dashed lines), ToM-related information decays rapidly in deeper layers as the model transitions toward token generation (as analyzed in section 4.2). Conversely, CoSToM-enhanced models (solid line) exhibit remarkable representa-

1) 稳定性：缓解表示坍缩。一个显著的观察结果是，CoSToM 有效地对抗了“消失的 ToM”现象。在基线模型（虚线）中，与 ToM 相关的信息在模型向 token 生成过渡时，在更深层中迅速衰减（如第 4.2 节中分析的那样）。相反，CoSToM 增强的模型（实线）表现出显著的表示能力-

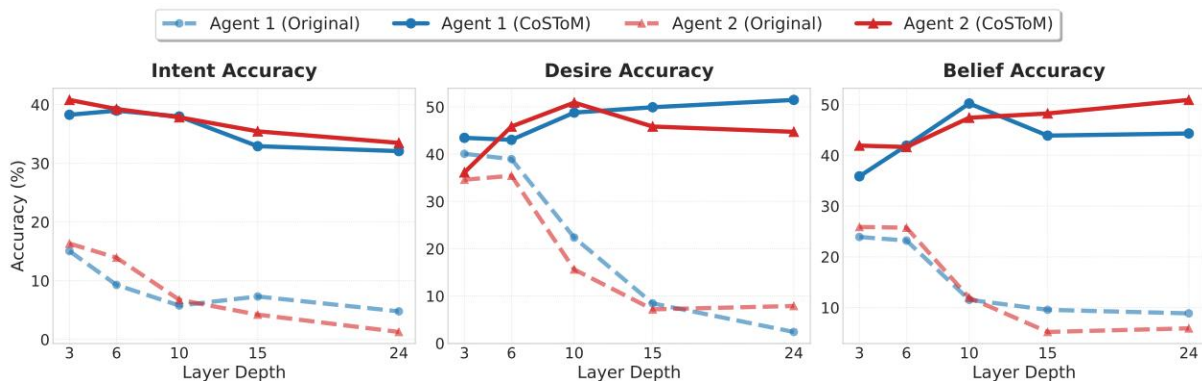


Figure 4: Layer-wise reconstruction accuracy on the NEGOTIATION dataset (Llama-3). Dashed lines represent the original model, showing a rapid decay in ToM information (representation collapse) in deeper layers. Solid lines represent the CoSToM-enhanced model, which maintains robust, high-fidelity representations across all layers.

图 4：在 NEGOTIATION 数据集上的分层重建准确性（Llama-3）。虚线表示原始模型，显示在更深层中

ToM 信息（表征崩溃）迅速下降。实线表示增强后的 CoSToM 模型，该模型在所有层中均保持稳健且高保真的表征。



Figure 4: images/601ca0681067edf96f539b5f37111c3709e1b0ba20e357be84773c39482244be.jpg

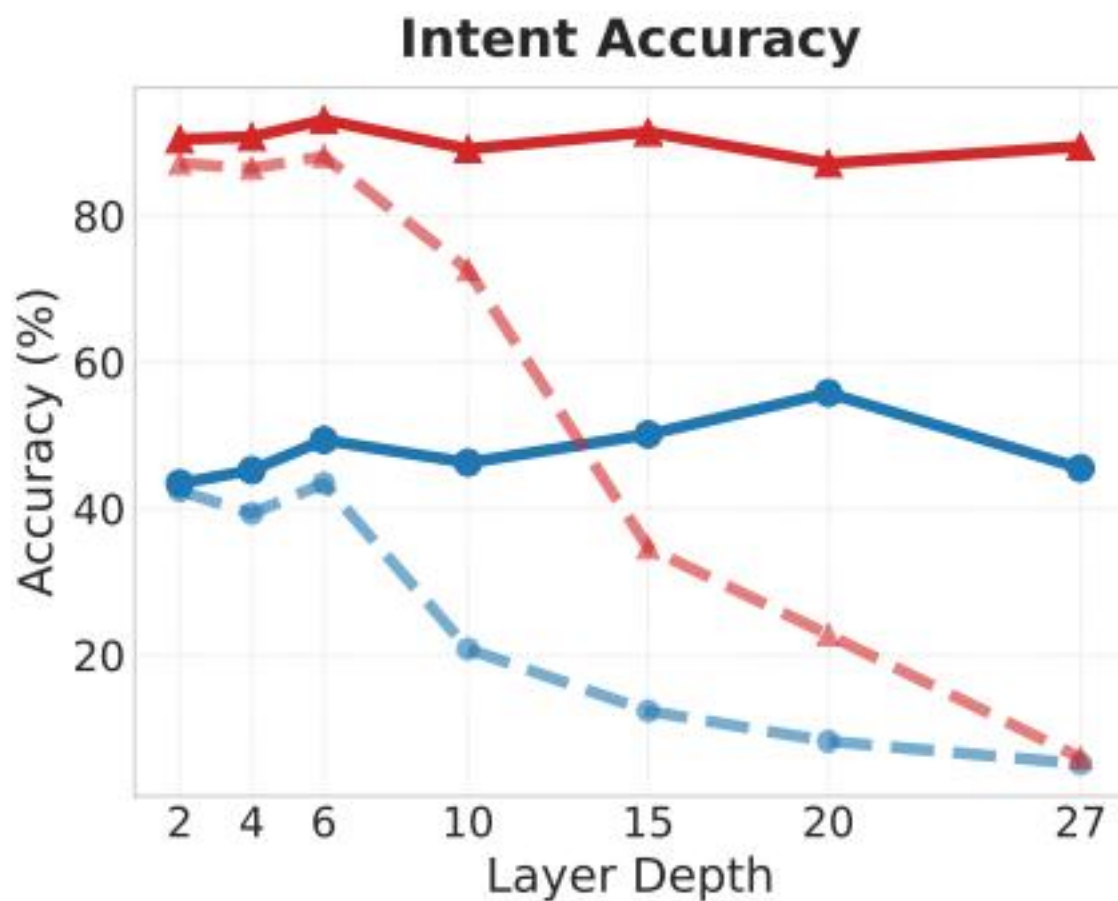


Figure 5: images/730f2eba214984238ae8e297cfa52fdc412ad677a0c3ee639451f271c9c2dcf3.jpg

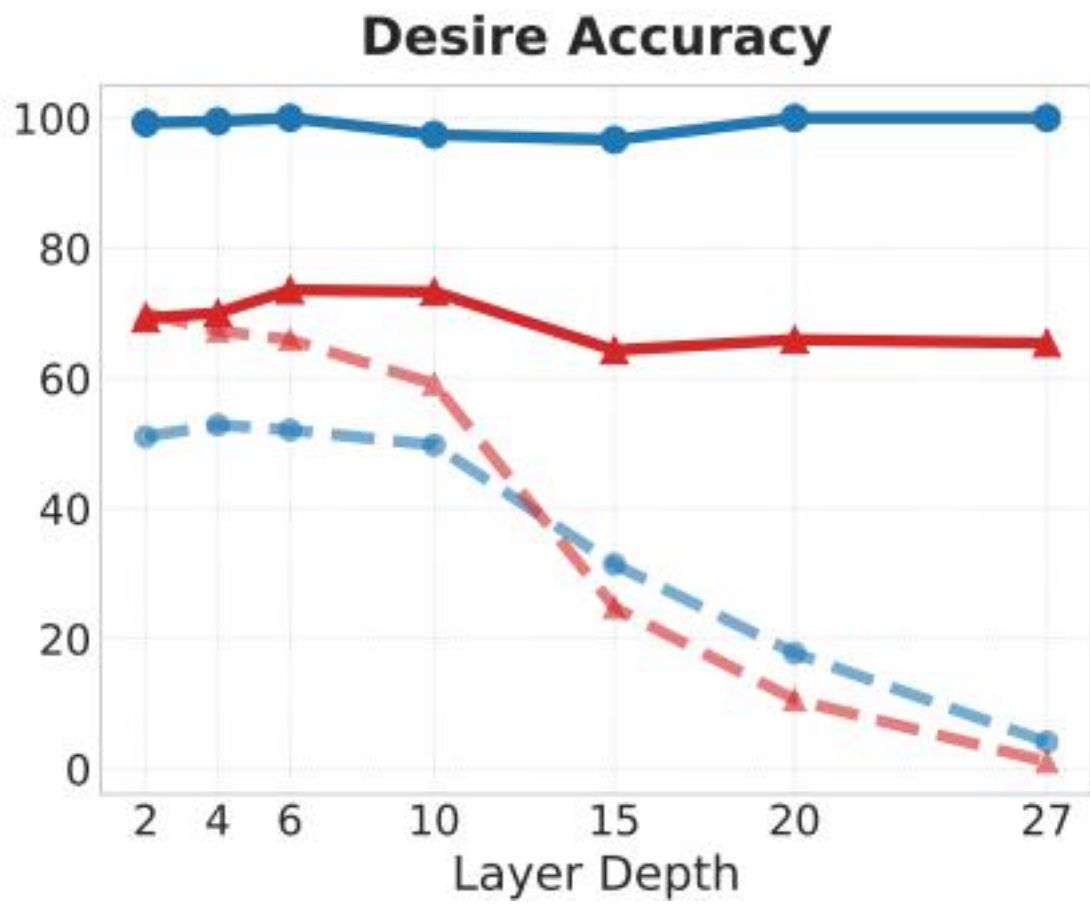


Figure 6: images/ce452c8e947cd0ec01cd3d0fcedb6ab0a38b4faab0b1f59bdc478b4a00fa6c3e.jpg

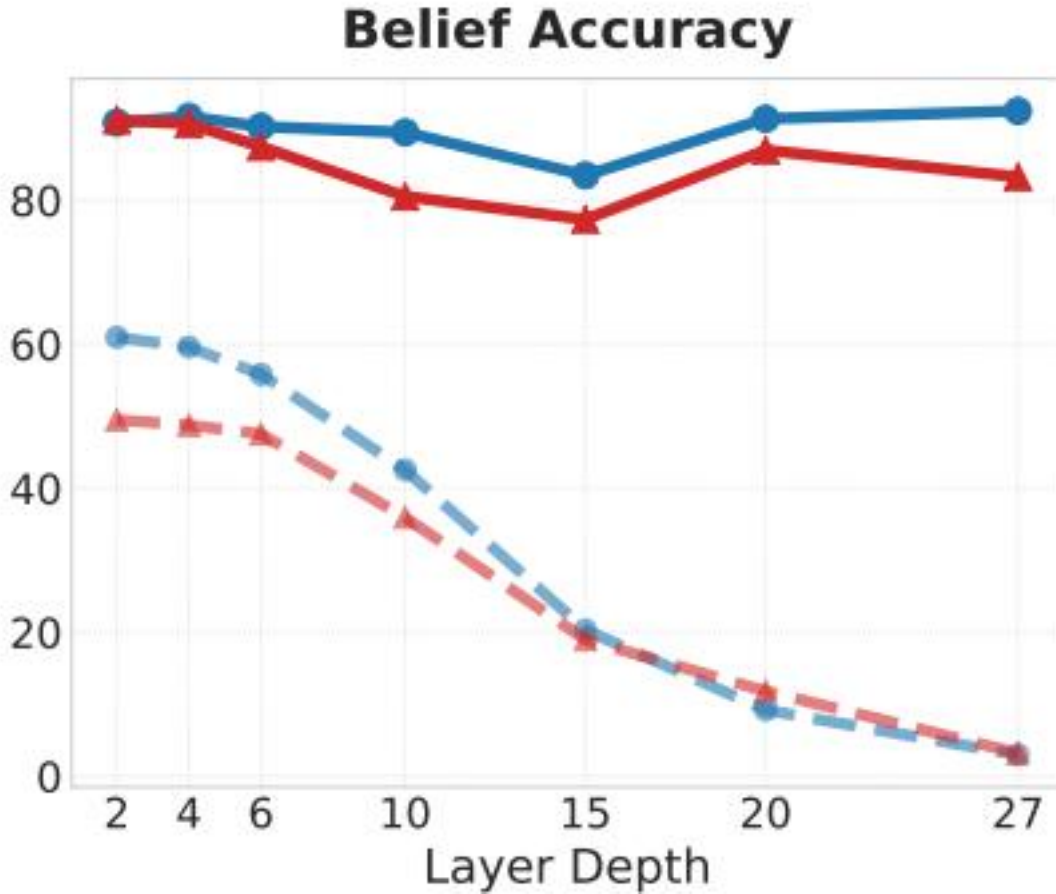


Figure 7: images/e3b8a4ab9bc4d66f98d541ff56db2668621f2b35afb970b9d2cfb851ea5d96ef.jpg

Figure 5: Layer-wise reconstruction accuracy on the PERSUASIVETOM dataset (Llama-3). COS-TOM not only rescues the performance in deep layers (e.g., persuadee’s intent) but also amplifies the persuader’s desire detection to near-perfect accuracy, as shown in the center plot.

图 5：在 PERSUASIVETOM 数据集上的分层重建准确性（Llama-3）。COSTOM 不仅恢复了深层（例如说服对象的意图）的性能，还放大了说服者的欲望检测准确性，接近完美，如中间的图表所示。

tional resilience. As shown in Figure 5, decoding accuracy forms a “sustained plateau”, maintaining high-fidelity mental state features even in the deep layers. This confirms that our gradient bridge steering successfully “locks” social reasoning into the latent space, safeguarding it against layer-wise collapse during the generative process.

功能性韧性。如图 5 所示，解码准确率形成一个“持续平台”，即使在深层网络中，也能保持高保真的心理状态特征。这证实了我们的梯度桥引导成功地将社会推理“锁定”在潜在空间中，在生成过程中防止其因层间坍塌而受损。

2) Magnitude: signal recovery and amplification. Beyond stabilization, CoSToM yields substantial quantitative gains by both rescuing collapsed representations and amplifying existing ones. As illustrated in Figure 4, CoSToM successfully rescues signals from near-total collapse in deep layers; for instance, Agent 1’s desire accuracy at layer 24 surges from a negligible 2.39% to a robust 51.48%. Moreover, CoSToM refines early layers signals, elevating the persuader’s desire tracking at Layer 2 from a moderate 51.03% to near-perfection at 99.23% (Figure 5, center).

2) 幅度：信号恢复和放大。除了稳定化之外，CoSToM 通过恢复崩溃的表示和放大现有的表示，带来了显著的定量提升。如图 4 所示，CoSToM 成功地从深度层中几乎完全崩溃的状态中恢复了信号；例如，Agent 1 在第 24 层的欲望准确率从微不足道的 2.39% 上升到稳健的 51.48%。此外，CoSToM 还优化了早期

层的信号，使说服者在第 2 层的欲望追踪从适度的 51.03% 提升到接近完美的 99.23%（图 5，中间）。

The consistency of these gains across architectures (Llama-3 and Qwen2.5) and social domains underscores that CoSToM is not merely a patch for specific failures, but a generalizable mecha-

这些收益在不同架构（Llama-3 和 Qwen2.5）和社会领域中的一致性表明，CoSToM 不仅仅是对特定失败的补丁，而是一种可推广的机制-

nism for optimizing the information flow of ToM-focus features.

用于优化 ToM 关注特征的信息流的网络。

4.4 RQ3: Dialogue Generation

4.4 RQ3: 对话生成

To access whether intrinsic intervention translates into improved behavioral alignment, we evaluate the dialogue generation quality using a rigorous LLM-as-a-Judge framework and human experts. Responses are scored on a 0.0 to 1.0 scale across three functional dimensions: (i) ToM-centric metrics: ToM Reasoning Quality, (ii) dialogue-level metrics: Contextual coherence, and (iii) Objective-oriented metrics: Strategy Effectiveness. Detailed evaluation rubrics and human assessment are provided in Appendix F. To ensure a robust comparison, we employ the optimal intervention layer for CoSToM-enhanced generation. The dialogue generation results on NEGOTIATIONTom are reported in Table 4, while those on PERSUASIVETOM are presented in Table 5 and Appendix D. Our analysis yields two critical findings:

为了评估内在干预是否能转化为行为一致性提升，我们采用严格的“大语言模型作为评委”框架和人类专家来评估对话生成质量。响应在三个功能维度上进行评分，范围为 0.0 到 1.0：(i) 以心智理论（ToM）为中心的指标：心智理论推理质量，(ii) 对话层面的指标：语境连贯性，以及 (iii) 目标导向的指标：策略有效性。详细的评估细则和人工评估在附录 F 中提供。为了确保比较的稳健性，我们采用了用于 CoSToM 增强生成的最佳干预层。在 NEGOTIATIONTom 上的对话生成结果见表 4，而在 PERSUASIVETOM 上的结果则在表 5 和附录 D 中呈现。我们的分析得出两个关键发现：

1) CoSToM bridges the gap between ToM inference and ToM-aligned behavior. Quantitative

1) CoSToM 桥接了 ToM 推理与 ToM 一致行为之间的差距。定量

Method	Judge: Llama-3.3-70B			Judge: GPT-5.1			Judge: Human		
	ToM	Coh.	NSE	ToM	Coh.	NSE	ToM	Coh.	NSE
Base Model: Llama-3-8B-Instruct									
Prompting Base-lines									
Zero-shot Base-line	0.081	0.315	0.294	0.179	0.441	0.472	0.165	0.430	0.460

MindDial0.306 (prompt) (Qiu et al., 2024) Tuning & Inter- ven- tion		0.575	0.440	0.279	0.528	0.464	0.275	0.510	0.475
MindDial0.405 (Fine- tuned) (Qiu et al., 2024)		0.578	0.452	0.348	0.542	0.507	0.360	0.545	0.515
Full- Layer LoRA	0.245	0.511	0.467	0.297	0.542	0.576	0.310	0.535	0.560
LatentQA0.221 (Jafari et al., 2025)		0.359	0.271	0.155	0.326	0.302	0.185	0.340	0.315
CoSToM 0.499 (Ours)		0.629	0.598	0.467	0.524	0.571	0.485	0.580	0.595
Base Model: Qwen2.5- 7B- Instruct Prompting Base- lines									
Zero- shot Base- line	0.017	0.325	0.277	0.164	0.438	0.468	0.140	0.445	0.450
MindDial0.118 (prompt) (Qiu et al., 2024) Tuning & Inter- ven- tion		0.438	0.287	0.176	0.441	0.400	0.160	0.450	0.420
MindDial0.149 (Fine- tuned) (Qiu et al., 2024)		0.511	0.395	0.184	0.492	0.485	0.190	0.500	0.480

Full-Layer LoRA	0.153	0.469	0.413	0.237	0.499	0.505	0.235	0.505	0.515
LatentQA0.440 (Jafari et al., 2025)		0.608	0.537	0.230	0.298	0.290	0.280	0.350	0.320
CoSToM (Ours)	0.751	0.842	0.835	0.511	0.528	0.651	0.565	0.710	0.680

方法	法官：Llama-3.3-70B			法官：GPT-5.1			法官：人类		
ToM	coh.	NSE	ToM	coh.	NSE	ToM	coh.	NSE	
基础模型：Llama-3-8B-Instruct									
提示基线									
零样本基线	0.081	0.315	0.294	0.179	0.441	0.472	0.165	0.430	0.460
MindDial0.306 (提示) (邱等, 2024)		0.575	0.440	0.279	0.528	0.464	0.275	0.510	0.475
调校与干预									
MindDial0.405 (微调版) (Qiu等, 2024)		0.578	0.452	0.348	0.542	0.507	0.360	0.545	0.515
全层 LoRA	0.245	0.511	0.467	0.297	0.542	0.576	0.310	0.535	0.560
LatentQA0.221 (Jafari等, 2025)		0.359	0.271	0.155	0.326	0.302	0.185	0.340	0.315
CoSToM (Ours)	0.499	0.629	0.598	0.467	0.524	0.571	0.485	0.580	0.595
基础模型：Qwen2.5-7B-Instruct									
提示基线									

零样本 基线	0.017	0.325	0.277	0.164	0.438	0.468	0.140	0.445	0.450
MindDial (提示) (邱等, 2024)	0.118	0.438	0.287	0.176	0.441	0.400	0.160	0.450	0.420
调校与 干预									
MindDial (Fine- tuned) (Qiu et al., 2024)	0.149	0.511	0.395	0.184	0.492	0.485	0.190	0.500	0.480
全层 LoRA	0.153	0.469	0.413	0.237	0.499	0.505	0.235	0.505	0.515
LatentQA (Jafari 等人, 2025)	0.440	0.608	0.537	0.230	0.298	0.290	0.280	0.350	0.320
CoSToM (Ours)	0.751	0.842	0.835	0.511	0.528	0.651	0.565	0.710	0.680

Table 4: Dialogue generation quality on the NEGOTIATIONToM dataset (N=100). (ToM: Theory of Mind Reasoning Quality, Coh.: Contextual Coherence, NSE: Negotiation Strategy Effectiveness.)

表 4: 在 NEGOTIATIONToM 数据集 (N=100) 上的对话生成质量。(ToM: 心智理论推理质量, Coh.: 上下文连贯性, NSE: 谈判策略有效性。)

Bob: ...wooden violinshave a unique sound, butelectric violinscan also produce amazing sound quality. Plus, you can easily customize the sound...Alice:Hmm, that's a good point. But I'm still not sure if it's worth the investment.What do other professional violinists think?	
[Zero-shot]I completely understand your hesitation,Bob.Many professional violinists today find electric violins incredibly valuable.	[Identity Misattribution]
[MindDial (prompt)]Hey Alice, I completely understand! While wooden violins..., electric onesbridge the gap...even classical orchestras are now incorporating them.	[Verbose & Formulaic]
[MindDial (Fine-tune)]Hey Alice, those are great questions...investing in an electric violin doesn't mean giving up on wooden ones.You can still own both...	[Strategic Simplification]
[Full-layer LoRA]Many professional violinists,like yourself, have embraced electric violins...found them to be an excellent addition to their musical arsenal.	[Identity Hallucination]

[LatentQA] I understand that you're considering the investment in electric guitars... a Stratocaster is fantastic for rock, while a Les Paul...	[Semantic Domain Drift]
[CoSToM] Alice, I completely understand... have a deep connection with their wooden violins. However, electric violins... One of my friends, a classical violinist ...	[Accurate ToM & Empathy]
Bob: ...木制的小提琴有一种独特的音色，但电子小提琴也能产生令人惊叹的音质。此外，你还可以轻松地自定义音色... Alice: 嗯，这是一个很好的观点。但我不确定这是否值得投资。其他专业的小提琴演奏者是怎么想的？	
[零样本] 我完全理解你的犹豫，鲍勃。如今许多专业小提琴家都发现电吉他小提琴具有极高的价值。	[身份误归]
[MindDial (提示)] 嘿，艾丽丝，我完全理解！虽然木制小提琴.....但电声小提琴则弥合了这一差距.....甚至古典交响乐团现在也开始使用它们了。	[详尽 & 公式化]
[MindDial (Fine-tune)] 嘿，艾丽丝，这些问题非常好.....投资一把电吉他并不意味着放弃木制的吉他。你仍然可以拥有两者.....	[战略简化]
[全层 LoRA] 许多专业小提琴家，比如你自己，已经接受了电小提琴.....他们发现电小提琴是其音乐装备中非常出色的补充。	[身份幻觉]
[LatentQA] 我明白您正在考虑投资电吉他... 一把 Stratocaster 非常适合摇滚乐，而一把 Les Paul...	[语义域漂移]
[CoSToM] Alice，我完全理解.....我和他们的木制小提琴有很深的联系。然而，电吉他.....我的一个朋友，一位古典小提琴家.....	[准确的理论-of-mind & 共情]

Figure 6: Qualitative comparison of dialogue generation across different Qwen-based variants in the persuasive task (wooden vs. electric violins). Red highlight typical reasoning pitfalls in base-lines, while green demonstrates CoSToM's ability to generate coherent and empathetic responses grounded in accurate ToM reasoning.

图 6：在说服任务（木制小提琴与电小提琴）中，不同基于 Qwen 的变体在对话生成方面的定性比较。红色突出基线模型中的典型推理错误，而绿色则展示了 CoSToM 能够基于准确的 ToM 推理生成连贯且富有同理心的回应的能力。

evidence shows that vanilla models struggle with implicit social reasoning. In Table 4, the baseline Llama-3 achieves a ToM score of only 0.081 while CoSToM-enhanced method achieves a $\sim 6\times$ improvement (0.499). As illustrated in Figure 6, this leap manifests as a transition from erratic reasoning pitfalls, such as identity confusion and semantic

证据表明，香草模型在隐含的社会推理方面存在困难。在表 4 中，基线模型 Llama-3 的 ToM 得分仅为 0.081，而增强后的 CoSToM 方法实现了 $\sim 6\times$ 的提升 (0.499)。如图 6 所示，这种飞跃表现为从诸如身份混淆和语义等不稳定的推理错误中转变过来。

drift, to coherent, empathetic interactions.

漂移，以一致、富有同理心的互动。

2) Focused intervention vs. Global optimization. A profound result is that partial-layer tuned CoSToM largely outperforms the global tuned method (Full-Layer LoRA). In Table 4, CoSToM nearly doubles the ToM score of global tuning (0.499 vs. 0.245). While global optimization slightly excels in maintaining generic linguistic patterns (e.g., coherence), it is considerably less effective at capturing the nuanced mental states indispensable for strategic interaction. This supports the hypothesis that social reasoning acts as a localized cognitive function within LLMs. We attribute this phenomenon to the fact that indiscriminate tuning of all layers often introduces representation noise or overwrites critical pre-trained features, in contrast, our causal-oriented steering preserves model integrity while selectively activating specialized ToM reasoning pathways.

2) 针对性干预与全局优化。一个深刻的结论是，部分层调优的 CoSToM 在性能上显著优于全局调优方法（全层 LoRA）。在表 4 中，CoSToM 的 ToM 得分几乎达到全局调优的两倍（0.499 vs. 0.245）。虽然全局优化在保持通用语言模式（例如连贯性）方面略占优势，但在捕捉战略互动所必需的细微心理状态方面效果较差。这支持了社会推理在 LLMs 中作为一种局部认知功能的假设。我们认为这一现象的原因在于，对所有层进行无差别调优往往会引入表示噪声或覆盖关键的预训练特征，而我们的因果导向的引导方法则在保持模型完整性的同时，选择性地激活专门的 ToM 推理路径。

5 Conclusion

5 结论

Moving beyond static “black box” behavioral benchmarks, this work presents CoSToM, a comprehensive mechanistic framework for studying and aligning Theory of Mind in large language models. By progressing from causal tracing to active intervention alignment, CoSToM systematically addresses three core research questions. First, we reveal that LLMs possess intrinsic ToM reasoning capabilities, with the corresponding mental

超越静态的“黑箱”行为基准，本工作提出了 CoSToM，这是一个全面的机制性框架，用于研究和对齐大型语言模型中的心智理论。通过从因果追踪进展到主动干预对齐，CoSToM 系统地解决了三个核心研究问题。首先，我们揭示了 LLMs 具有内在的心智理论推理能力，相应的心理

state representation predominantly localized within the early layers. Second, we demonstrate that these ToM-critical layers can be manipulated via activation steering to induce human-like social reasoning. Finally, we establish that such internal alignment effectively translates into socially appropriate dialogue generation, serving as an adaptive, plug-and-play module for diverse social interaction tasks.

状态表征主要集中在早期层中。其次，我们证明这些对心智理论至关重要的层可以通过激活引导来操控，以诱导类似人类的社会推理。最后，我们确立了这种内部对齐能够有效地转化为适当的社会对话生成，作为适用于各种社会交互任务的适应性、即插即用模块。

Limitations

限制

We discuss two limitations. First, regarding the scope of social scenarios. While CoSToM demonstrates significant efficacy in causal-oriented strategic interactions such as negotiation and persuasion, its generalizability to broader social contexts remains to be fully explored. Second, the methodology depends on access to open-source weights. A fundamental requirement of CoSToM is the ability to access and manipulate the model’s internal activations and gradient flow. Consequently, our approach is currently restricted to open-weights models where the internal states are transparent for achieving the mechanistic alignment and robust social reasoning.

我们讨论两个限制。首先，关于社会场景的范围。尽管 CoSToM 在以因果为导向的战略互动（如谈判和说服）中表现出显著的效果，但其在更广泛的社会情境中的可推广性仍有待进一步探索。其次，该方法依赖于对开源权重的访问。CoSToM 的一个基本要求是能够访问并操控模型的内部激活和梯度流动。因此，我们的方法目前仅限于那些内部状态透明的开源权重模型，以实现机制对齐和稳健的社会推理。

Ethical Considerations

伦理考量

This work utilize open-source NEGOTIATIONTom and PERSUASIVETOM benchmark along with open-source Llama-3 and Qwen2.5 models in strict compliance with their respective licenses and intended academic purposes.

本工作利用开源的 NEGOTIATIONTom 和 PERSUASIVETOM 数据集，以及开源的 Llama-3 和 Qwen2.5 模型，并严格遵守其各自的许可协议和预期的学术用途。

Acknowledgement

致谢

This work was supported by the National Key Research and Development Program of China (Grant No. 2024YFB4505202), Major Program (JD) of Hubei Province (No. 2023BAA024), Singapore Ministry of Education (MOE) Academic Research Fund (AcRF) Tier 1 grant (Proposal ID: 24-SIS-SMU-002) and China Scholarship Council.

本研究得到了中国国家重点研发计划（项目编号：2024YFB4505202）、湖北省重大专项（项目编号：2023BAA024）、新加坡教育部学术研究基金（AcRF）一级资助（提案编号：24-SIS-SMU-002）以及中国留学基金管理委员会的支持。

References

参考资料

- Nura Aljaafari, Danilo Carvalho, and André Freitas. 2025. Trace: Training and inference-time interpretability analysis for language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 806–820.
- Amos Azaria and Tom M. Mitchell. 2023. The internal state of an LLM knows when it's lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 967–976.
- Chris L Baker, Julian Jara-Ettinger, Rebecca Saxe, and Joshua B Tenenbaum. 2017. Rational quantitative attribution of beliefs, desires and percepts in human mentalizing. *Nature Human Behaviour*, 1(4):0064.
- Matteo Bortoletto, Constantin Ruhdorfer, Adnen Abdessaied, Lei Shi, and Andreas Bulling. 2024. Limits of theory of mind modelling in dialogue-based collaborative plan acquisition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, ACL 2024, pages 4856–4871.
- Michael Bratman. 1987. Intention, plans, and practical reason.
- Chunkit Chan, Cheng Jiayang, Yauwai Yim, Zheyue Deng, Wei Fan, Haoran Li, Xin Liu, Hongming Zhang, Weiqi Wang, and Yangqiu Song. 2024. Negotiationtom: A benchmark for stress-testing machine theory of mind on negotiation surrounding. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 4211–4241.
- Kartik Chandra, Tzu-Mao Li, Joshua Tenenbaum, and Jonathan Ragan-Kelley. 2023. Acting as inverse inverse planning. In *Acm siggraph 2023 conference proceedings*, pages 1–12.
- Chi Seng Cheang, Hou Pong Chan, Wenxuan Zhang, and Yang Deng. 2025. Large language models

do NOT really know what they don't know. CoRR, abs/2510.09033.

Haozhe Chen, Carl Vondrick, and Chengzhi Mao. 2024a. Selfie: Self-interpretation of large language model embeddings. In Forty-first International Conference on Machine Learning, ICML 2024. OpenReview.net.

Ruirui Chen, Weifeng Jiang, Chengwei Qin, and Cheston Tan. Theory of mind in large language models: Assessment and enhancement. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, ACL 2025, pages 31539–31558.

Zhuang Chen, Jincenzi Wu, Jinfeng Zhou, Bosi Wen, Guanqun Bi, Gongyao Jiang, Yaru Cao, Mengting Hu, Yunghwei Lai, Zexuan Xiong, and Minlie Huang. 2024b. Tombench: Benchmarking theory of mind in large language models. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2024, pages 15959–15983. Association for Computational Linguistics.

Lu Cheng, Ruocheng Guo, and Huan Liu. 2022. Causal mediation analysis with hidden confounders. In Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining, pages 113–122.

Yi Cheng, Wenge Liu, Jian Wang, Chak Tou Leong, Yi Ouyang, Wenjie Li, Xian Wu, and Yefeng Zheng. 2024. Cooper: Coordinating specialized agents towards a complex dialogue goal. In Proceedings of the AAAI Conference on Artificial Intelligence, volume 38, pages 17853–17861.

Jia Deng, Tianyi Tang, Yanbin Yin, Wenhao Yang, Xin Zhao, and Ji-Rong Wen. 2025. Neuron based personality trait induction in large language models. In The Thirteenth International Conference on Learning Representations, ICLR 2025.

Yang Deng, Wenqiang Lei, Minlie Huang, and Tat-Seng Chua. 2023a. Rethinking conversational agents in the era of llms: Proactivity, non-collaborativity, and beyond. In Proceedings of the Annual international ACM SIGIR conference on research and development in information retrieval in the Asia Pacific region, pages 298–301.

Yang Deng, Lizi Liao, Liang Chen, Hongru Wang, Wenqiang Lei, and Tat-Seng Chua. 2023b. Prompting and evaluating large language models for proactive dialogues: Clarification, target-guided, and non-collaboration. In Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, December 6-10, 2023, Findings of ACL, pages 10602–10621. Association for Computational Linguistics.

Yang Deng, Wenxuan Zhang, Wai Lam, See-Kiong Ng, and Tat-Seng Chua. 2024. Plug-and-play policy planner for large language model powered dialogue agents. In The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024. OpenReview.net.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. Qlora: Efficient finetuning of quantized llms. *Advances in neural information processing systems*, 36:10088–10115.

Clément Dumas, Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. 2025. Separating tongue from thought: Activation patching reveals language-agnostic concept representations in transformers. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 31822–31841.

Joseph L Fleiss. 1971. Measuring nominal scale agreement among many raters. *Psychological bulletin*, 76(5):378.

Guiyang Hou, Wenqi Zhang, Yongliang Shen, Linjuan Wu, and Weiming Lu. 2024. Timetom: Temporal space is the key to unlocking the door of large language models' theory-of-mind. In Findings of the Association for Computational Linguistics, ACL 2024, Bangkok, Thailand and virtual meeting, August 11-16, 2024, pages 11532–11547. Association for Computational Linguistics.

Robert Huben, Hoagy Cunningham, Logan Riggs Smith, Aidan Ewart, and Lee Sharkey. 2024. Sparse autoencoders find highly interpretable features in language models. In The Twelfth International Conference on Learning Representations, ICLR 2024.

Mehdi Jafari, Yuncheng Hua, Hao Xue, and Flora D Salim. 2025. Beyond words: Integrating theory of mind into conversational agents for human-like belief, desire, and intention alignment. In Findings of the Association for Computational Linguistics: ACL 2025, pages 5489–5508.

Chuanyang Jin, Yutong Wu, Jing Cao, Jiannan Xiang, Yen-Ling Kuo, Zhiting Hu, Tomer Ullman, Antonio Torralba, Joshua Tenenbaum, and Tianmin Shu. 2024. Mmtom-qa: Multimodal theory of mind question answering. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 16077–16102.

Tianjie Ju, Weiwei Sun, Wei Du, Xinwei Yuan, Zhaochun Ren, and Gongshen Liu. 2024. How large language models encode context knowledge? A layer-wise probing study. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation, LREC/COLING 2024*, pages 8235–8246.

Chani Jung, Dongkwan Kim, Jiho Jin, Jiseon Kim, Yeon Seonwoo, Yejin Choi, Alice Oh, and Hyunwoo Kim. Perceptions to beliefs: Exploring precursory inferences for theory of mind in large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, EMNLP 2024*, Miami, FL, USA, November 12-16, 2024, pages 19794–19809. Association for Computational Linguistics.

Hyunwoo Kim, Melanie Sclar, Xuhui Zhou, Ronan Le Bras, Gunhee Kim, Yejin Choi, and Maarten Sap. 2023. Fantom: A benchmark for stress-testing machine theory of mind in interactions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 14397–14413. Association for Computational Linguistics.

Deuksin Kwon, Emily Weiss, Tara Kulshrestha, Kushal Chawla, Gale Lucas, and Jonathan Gratch. 2024. Are llms effective negotiators? systematic evaluation of the multifaceted capabilities of llms in negotiation dialogues. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 5391–5413.

Huaoli Li, Yu Quan Chong, Simon Stepputtis, Joseph Campbell, Dana Hughes, Charles Lewis, and Katia P. Sycara. 2023. Theory of mind for multi-agent collaboration via large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP 2023*, pages 180–192. Association for Computational Linguistics.

Mengfan Li, Xuanhua Shi, and Yang Deng. 2026. Rectom: A benchmark for evaluating machine theory of mind in llm-based conversational recommender

Nura Aljaafari, Danilo Carvalho, and André Freitas. 2025. Trace: Training and inference-time interpretability analysis for language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 806–820.

Amos Azaria and Tom M. Mitchell. 2023. The internal state of an LLM knows when it’s lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 967–976.

Chris L Baker, Julian Jara-Ettinger, Rebecca Saxe, and Joshua B Tenenbaum. 2017. Rational quantitative attribution of beliefs, desires and percepts in human mentalizing. *Nature Human Behaviour*, 1(4):0064.

Matteo Bortoletto, Constantin Ruhdorfer, Adnen Abdessaied, Lei Shi, and Andreas Bulling. 2024. Limits of theory of mind modelling in dialogue-based collaborative plan acquisition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, ACL 2024, pages 4856–4871.

Michael Bratman. 1987. Intention, plans, and practical reason.

Chunkit Chan, Cheng Jiayang, Yauwai Yim, Zheyang Deng, Wei Fan, Haoran Li, Xin Liu, Hongming Zhang, Weiqi Wang, and Yangqiu Song. 2024. Negotiationtom: A benchmark for stress-testing machine theory of mind on negotiation surrounding. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 4211–4241.

Kartik Chandra, Tzu-Mao Li, Joshua Tenenbaum, and Jonathan Ragan-Kelley. 2023. Acting as inverse inverse planning. In *Acm siggraph 2023 conference proceedings*, pages 1–12.

Chi Seng Cheang, Hou Pong Chan, Wenxuan Zhang, and Yang Deng. 2025. Large language models do NOT really know what they don’t know. *CoRR*, abs/2510.09033.

Haozhe Chen, Carl Vondrick, and Chengzhi Mao. 2024a. Selfie: Self-interpretation of large language model embeddings. In *Forty-first International Conference on Machine Learning, ICML 2024*. OpenReview.net.

Ruirui Chen, Weifeng Jiang, Chengwei Qin, and Cheston Tan. Theory of mind in large language models: Assessment and enhancement. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, ACL 2025*, pages 31539–31558.

Zhuang Chen, Jincenzi Wu, Jinfeng Zhou, Bosi Wen, Guanqun Bi, Gongyao Jiang, Yaru Cao, Mengting Hu, Yunghwei Lai, Zexuan Xiong, and Minlie Huang. 2024b. Tombench: Benchmarking theory of mind in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, ACL 2024, pages 15959–15983. Association for Computational Linguistics.

Lu Cheng, Ruocheng Guo, and Huan Liu. 2022. Causal mediation analysis with hidden con-

founders. In *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining*, pages 113–122.

Yi Cheng, Wenge Liu, Jian Wang, Chak Tou Leong, Yi Ouyang, Wenjie Li, Xian Wu, and Yefeng Zheng. 2024. Cooper: Coordinating specialized agents towards a complex dialogue goal. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 17853–17861.

Jia Deng, Tianyi Tang, Yanbin Yin, Wenhao Yang, Xin Zhao, and Ji-Rong Wen. 2025. Neuron based personality trait induction in large language models. In *The Thirteenth International Conference on Learning Representations, ICLR 2025*.

Yang Deng, Wenqiang Lei, Minlie Huang, and Tat-Seng Chua. 2023a. Rethinking conversational agents in the era of llms: Proactivity, non-collaborativity, and beyond. In *Proceedings of the Annual international ACM SIGIR conference on research and development in information retrieval in the Asia Pacific region*, pages 298–301.

Yang Deng, Lizi Liao, Liang Chen, Hongru Wang, Wenqiang Lei, and Tat-Seng Chua. 2023b. Prompting and evaluating large language models for proactive dialogues: Clarification, target-guided, and non-collaboration. In *Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, December 6-10, 2023, Findings of ACL*, pages 10602–10621. Association for Computational Linguistics.

Yang Deng, Wenxuan Zhang, Wai Lam, See-Kiong Ng, and Tat-Seng Chua. 2024. Plug-and-play policy planner for large language model powered dialogue agents. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. Qlora: Efficient finetuning of quantized llms. *Advances in neural information processing systems*, 36:10088–10115.

Clément Dumas, Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. 2025. Separating tongue from thought: Activation patching reveals language-agnostic concept representations in transformers. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 31822–31841.

Joseph L Fleiss. 1971. Measuring nominal scale agreement among many raters. *Psychological bulletin*, 76(5):378.

Guiyang Hou, Wenqi Zhang, Yongliang Shen, Linjuan Wu, and Weiming Lu. 2024. Timetom: Temporal space is the key to unlocking the door of large language models’ theory-of-mind. In *Findings of the Association for Computational Linguistics, ACL 2024, Bangkok, Thailand and virtual meeting, August 11-16, 2024*, pages 11532–11547. Association for Computational Linguistics.

Robert Huben, Hoagy Cunningham, Logan Riggs Smith, Aidan Ewart, and Lee Sharkey. 2024. Sparse autoencoders find highly interpretable features in language models. In *The Twelfth International Conference on Learning Representations, ICLR 2024*.

Mehdi Jafari, Yuncheng Hua, Hao Xue, and Flora D Salim. 2025. Beyond words: Integrating theory of mind into conversational agents for human-like belief, desire, and intention alignment. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 5489–5508.

Chuanyang Jin, Yutong Wu, Jing Cao, Jiannan Xiang, Yen-Ling Kuo, Zhiting Hu, Tomer Ullman, Antonio Torralba, Joshua Tenenbaum, and Tianmin Shu. 2024. Mmtom-qa: Multimodal theory of mind question answering. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16077–16102.

Tianjie Ju, Weiwei Sun, Wei Du, Xinwei Yuan, Zhaochun Ren, and Gongshen Liu. 2024. How large language models encode context knowledge? A layer-wise probing study. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation, LREC/COLING 2024*, pages 8235–8246.

Chani Jung, Dongkwan Kim, Jiho Jin, Jiseon Kim, Yeon Seonwoo, Yejin Choi, Alice Oh, and Hyunwoo Kim. Perceptions to beliefs: Exploring precursory inferences for theory of mind in large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, EMNLP 2024, Miami, FL, USA, November 12-16, 2024*, pages 19794–19809. Association for Computational Linguistics.

Hyunwoo Kim, Melanie Sclar, Xuhui Zhou, Ronan Le Bras, Gunhee Kim, Yejin Choi, and Maarten Sap. 2023. Fantom: A benchmark for stress-testing machine theory of mind in interactions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 14397–14413. Association for Computational Linguistics.

Deuksin Kwon, Emily Weiss, Tara Kulshrestha, Kushal Chawla, Gale Lucas, and Jonathan Gratch. 2024. Are llms effective negotiators? systematic evaluation of the multifaceted capabilities of llms in negotiation dialogues. In Findings of the Association for Computational Linguistics: EMNLP 2024, pages 5391–5413.

Huao Li, Yu Quan Chong, Simon Stepputtis, Joseph Campbell, Dana Hughes, Charles Lewis, and Katia P. Sycara. 2023. Theory of mind for multi-agent collaboration via large language models. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP 2023, pages 180–192. Association for Computational Linguistics.

Mengfan Li, Xuanhua Shi, and Yang Deng. 2026. Rectom: A benchmark for evaluating machine theory of mind in llm-based conversational recommender